Machine Learning

- Machine Learning always do the "prediction" for data is required.
- Based on past data we will predict the data
- The Basic resources for ML are UCI, Kaggle (or) Create own dataset.
- Data science is about Collecting, cleaning, analyzing & visualizing data to generate insights.

Machine Learning :- ML empowers this process by providing predictive and automated tools to ~~empower~~ model patterns in data. Ex:- Rain prediction, Top website (Ranking Algorithm).

Career opportunities :-

Roles of ML like :

- Data scientist
- ML Engineer
- AI/ML Researcher

Roadmap for ML Engineer :-

SKILLS	what to learn
Maths	Linear Algebra, probability, Statistics, calculu
Programming	Python (Numpy, Pandas, matplotlib), oop
Data structures & Algorithms	Arrays, Linked Lists, Trees, Graphs, DP.

- Master Data pipelines & Deployment (Docker, Fast API, Flask, GIT).
- Do Internships & freelance
  - Try platforms like Turing, Linkedin, upwork, Internshala
  - Apply for : ML Intern / AI Intern / Data Analyst Intern.

Human learning :-

Human Learning based on "Experience".

- Human Learning is a natural cognitive process through which individuals acquire new knowledge, develop skills, modify behaviours, and gain understanding through experience, study, instruction, observation.

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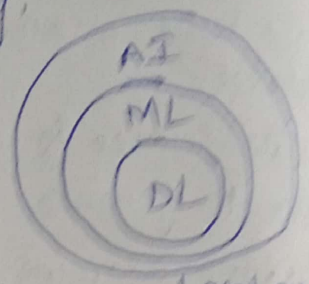
Key characteristics of Human Learning :-

i) Experience - based
ii) Adaptive
iii) Generalisation
iv) Memory integration
v) Social Learning.



Types of Human Learning :-

i) Learning with Guidance
ii) Self - directed Learning
iii) Learning through consequences.

Types of ML :-

i) Supervised - labelled data based on regression & classification (continuous) based on label ex- stock market prediction, weather, ecological data

ii) unSupervised - Unlabelled data (grouping) apply cluster algorithm.

iii) Reinforcement - learned from the past experience. algorithm - Q learning. ex- Self drive cars, online chess game

What is Machine Learning ?

ML is the Subset of AI that enables Computer Systems to automatically learn ~~and~~ improve performance on a specific task through experience without being explicitly programmed for every possible scenario.

Principles of ML :-

i) Data Driven : Algorithms learn from data rather than pre-programmed rules.

ii) pattern Recognition : Identification of patterns & relationships in data

iii) Generalization : Ability to make accurate predictions on unseen data.

iv) Automatic improvement

v) Adaptation.

Linear Regression ⇒ $y = mx + c$ (y: Price, x: Independent)

* i/p + program = o/p ⇒ for all programming languages.

* But in ML, i/p + o/p = model.

x_1, x_2, x_3 → features, y → only one.

model - ready to use anyone.

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Steps in ML :- Data collection

↓

Data preprocessing (To remove duplicates, null values, empty columns, unwanted features).

↓

Model selection - Based on the data i.e; structured, unstructured, reinforcement.

↓

Training & Testing - we apply 80 - 20 method. Training Testing

↓

Evaluation - Based on accuracy, sensitivity, F1 score, specificity.

↓

Model → finally it predicts.

Numerical, categorical, labelled, classification & regression, numerical, unlabelled, clustering

when we don't get accurate value

* Regression :-

- To predict the values continuously i.e; numerical values.
- There are two types of Regression:
  - i) Linear Regression
  - ii) Multi variate Regression (or) Linear with multiple variable

i) Linear Regression with one variable (x) :-

$$\boxed{y = mx + b}$$

- "Model" is nothing but ready to use anyone.
- Regression comes under linear model.
- Linear model finds the less error rate when you applied linear regression.
- Regression finds the relationship b/w dependent (o/p) & Independent variables. (Input)

$y = mx + b$ → y-intercept.

m → Slope (or) Gradient → reduce the error rate.

Code :-

```
import Pandas as pd
import numpy as np
from sklearn import linear_model
import matplotlib.pyplot as plt
df = pd.read_csv('houseprice.csv')
df
```

Create Linear Regression object

reg = linear_model.LinearRegression()

reg.fit(new_df, price)

LinearRegression(copy_X = True, fit_intercept = True, n_jobs = None, Normalize = false).

i) predict price of a home with area = 3300 sq ft

reg.predict([[3300]])

array([628715.7534 2466])

reg.coef_

array([135.78])

Linear Regression with Multiple variables:-

- when there are any missing values and fill that value with mean (or) median.

$$\boxed{y = m_1x_1 + m_2x_2 + m_3x_3 + b.}$$

price = m_1 * area + m_2 * bedrooms + m_3 * age + b.

dependent

Independent

Code:-

```
import pandas as pd
import numpy as np
from sklearn import linear_model
df = pd.read_csv('homeprices.csv')
df
```

	area	bedrooms	age	price
0	2600	3.0	20	550000
1	3000	4.0	15	565000
2	3200	NAN	18	610000
3	3600	3.0	30	595000
4	4000	5.0	8	760000
5	4100	6.0	8	810000

Data preprocessing : Fill NA values with median value of a colum

```
df.bedrooms.median()
```

4.0

df.bedrooms = df.bedrooms.fillna(df.bedrooms.median())

df

	area	bedrooms	age	price
0	2600	3.0	20	550000
1	3000	4.0	15	565000
2	3200	4.0	18	610000
3	3600	3.0	30	595000
4	4000	5.0	8	760000
5	4100	6.0	8	810000

reg = linear_model.LinearRegression()

reg.fit(df.drop('price', axis='columns'), df['price'])

LinearRegression(copy_X=True, fit_intercept=True, n_jobs=None, normalize=False)

reg.coef_

array([112.06244194, 23388.88007794, -3231.71790863])

reg.intercept_

221323.00186540408

Find price of home with 3000 sqr ft area, 3 bedrooms, 40 year old

reg.predict([[3000, 3, 40]])

array([498408.25158031])

Model Selection

Training: We will train our model on this dataset (.fit

Testing: We will use this subset to make actual Predictions using trained model.

from sklearn.model_selection import train_test_split

X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.3)

X_train

Identify the accuracy by using score.

Tensorflow loss value examples:-

- Sparse_categorical_crossentropy
- Mean absolute error (MAE)
- Mean square error. (MSE)

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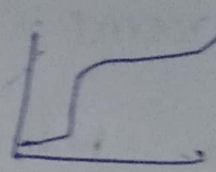
Linear Regression : $y = mx + c$

Logistic Regression : $\boxed{y = \frac{1}{e^{-(mx+c)}}}$



↳ It is used to show the loss and errors. in the form of binary values (0's & 1's).

* Bias & weights are random in the range of 0 to 1.

↳ error.

$$\boxed{\text{Loss (or) error} = \text{Actual values} - \text{predicted values}}$$

Total error:- for each sample we have to find the error rate. The sum of all error rates is called Total error.

$$\text{Total error} = \text{error}_1 + \text{error}_2 + \dots + \text{error}_{13} \text{ (loss)}$$

$$= \sum_{i=1}^{n} abs(y_i - \hat{y}_i)$$

actual predicted

$$\text{Mean absolute error (MAE)} = \frac{1}{n} \sum_{i=1}^{n} abs(y_i - \hat{y}_i)$$

(cost function)

- Initially weights & bias are random but the model will updated the values when it reaches to "Convergence" (previous and present weights are equal).

$$\Rightarrow \text{Mean squared error (MSE)} = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

(cost function)

Overfitting and underfitting in ML:-

The model will cross accuracy 1.0 it is overfitting.

Bias is nothing but error. [actual value - predicted value].

- $\boxed{\text{Total error} = \text{bias} + \text{variance}}$

bias = e1+e2+e3/3;

Variance = (e1+b)² + (e2+b)² + (e3+b)²

*Overfitting occurs when "low bias and high variance"

why it comes ↳ starts caching noise & inaccurate values present in the dataset.

- If model is overfitted it reduces the efficiency.

How to avoid overfitting :-

- using k-fold cross validation (dataset into partitions and use it training & testing k-fold)

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* Paradigms of ML :-

i) learning by Rote (or) memorization

ii) learning by deduction

(logical inference) Ex: if A is true then B is also true in no

iii) learning by abduction. $A \to B$ Ex: aeroplane.

Ex: If A is true then B may or maynot be true in

iv) learning by induction Ex: bird

- Supervised learning → learning from Examples → label name is required.
- unsupervised learning → learning from observations.

⇒ Same type of data is grouping. There is no need of label name. → 0 contains many types, we can identif based on label name.

v) learning by interaction & rewards

↳ Reinforcement learning - learn from the past Experienc

* Matching :-

- Compare ~~any~~ of the ~~metrics~~ vectors either similar or dissimilar.

i) Distance measure / Dissimilarity measure

- Using Euclidean distance : $d(u,v) = \sqrt{\sum_{i=1}^{l} (u_i - v_i)^2}$
- Manhattan distance : $d(u,v) = \sum_{i=1}^{l} |u_i - v_i|$

Ex:- $u = (1,2)$
$v = (4,6)$

E: $d = \sqrt{(1-4)^2 + (2-6)^2} = \sqrt{9+16} = \sqrt{25} = 5$

M: $d = |(1-4)+(2-6)| = |3+4| = 7$.

ii) Similarity measure

- Cosine Similarity : $\cos(u,v) = \frac{u^T v}{|u| \cdot |v|}$ (dot product of

Applications of ~~metha~~. matching :-

i) Nearest Neighbour Search. - find based on Euclidean distance

data: $x_1(1,1) \leftarrow$ cat $\quad d(x,x_1) = \sqrt{(1-3)^2+(1-4)^2} = \sqrt{13}$

$x_2(2,3) \leftarrow$ dog $\quad d(x,x_2) = \sqrt{(2-3)^2+(3-4)^2} = \sqrt{2}$

$x_3(4,5) \leftarrow$ dog $\quad d(x,x_3) = \sqrt{(4-3)^2+(5-4)^2} = \sqrt{2}$.

i/p :- (3,4) $\quad (x_2, x_3)$ is same so the nearest neighbour search is dog.

Exercises :-

1. You are given that 9x17 is 153 and 4x17 is 68. From this data, you need to learn the value of 13x17, which learning paradigm is useful here ?.

Sol:-

$$\begin{cases} 9 \times 17 = 153 \\ 4 \times 17 = 68 \end{cases} \qquad 13 \times 17 = ?$$

$$\begin{array}{r} 153 \\ 68 \\ \hline 221 \end{array}$$

$$(9+4) \cdot 17 = 153+68 = 221.$$

This is a "Supervised learning" [learning ~~from~~ by induction

2. You are given the following statements :

a) The sum of two even numbers is even.

b) 12 is an even number

c) 22 is an even number.

- learning by deduction (logical inference).

3. Consider the following statements :

a) If x is an even number, then x+1 is odd and x+2 is even

b) 34 is an even number.

which learning paradigm is used to learn that 37 is odd and 38 is even ?

- learning by deduction (logical inference).

4. Consider the following reasoning : a) If x is odd, then x+1 is

You have learnt from the above that 21 is odd. which learning paradigm is used ? Specify any

b) 22 is even.

- learning by deduction (logical inference).

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5. Consider the following attributes. find out whether they are nominal (categorical), ordinal & numerical assumptions.

range (median, ...)

label data (blood group, T (F))

↳ by using "one hot encoding" technique is used in ML.

↳ Ex: i) shirt sizes (shot, medium, large)

ii) rating (1-4)

↳ by using "label encoding" handles ML.

Ex: Survey Rating.

a) Telephone number

b) Feature that takes values from {ball, bat, wicket, umpire, batsman, bowler} 6x6 matrix

c) Temperature

d) weight

e) Feature that takes values from {short, medium, height, tall},

numerical :- Quantative

discrete, continuous.

handles ML is (z-score) standardization, (min max) normalisation

- used mean, variance and another statistical measures.

discrete - countable integers.

Continuous - Measurable real numbers

Ex: weight, temperature,

Feature Engg :-

⇒ nominal, ordinal, numerical assumptions all are comes under the "feature engineering."

one hot encoding, label encoding, normalisation, standardis notation.

a) Comes under nominal :- numbers are identifiers not quantified.

↳ fixed. Ex:- zip code, bank customer id.

b) 6x6 matrix (nominal).

c) Numerical d) Numerical e) ordinal.

6. Let x_i and x_j be two dimensional unit normal vectors; $||x_i||=1$ and $||x_j||=1$. Derive the relation b/w the Euclidean distance $d(x_i, x_j)$ and cosine of the angle b/w x_i and x_j.

Sol:- $|x|=1$, $|y|=1$

$$d(x,y)^2 = (|x|-|y|)^2$$

$$= |x|^2+|y|^2-2|x||y|$$

$$= 1+1-2(1)(1)$$

$$= 2-2\cos\theta.$$

$\boxed{x \cdot y = \cos\theta}$ dot product for unit vectors.

$$d(x,y)^2 = 2(1-\cos\theta)$$

$$d(x,y) = \sqrt{2(1-\cos\theta)}$$

$\theta = 0 \Rightarrow 1$

$\theta = 180 \Rightarrow -1$

$\theta = 90 \Rightarrow$

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7. Consider the dataset :
(1,1,1), (1,1,2), (1,1,3), (1,2,2), (1,1,–), (6,6,10).
↓ outlier.
Predict the missing value in the pattern vector (1,1,–) using KNNS with k=3. what happens when k=5?

Sol:- Assume – as "0".

$x_1 = \sqrt{(1-1)^2+(1-1)^2+(0-1)^2} = \sqrt{0+0+1} = \sqrt{1} = 1$

$x_2 = \sqrt{(1-1)^2+(1-1)^2+(0-2)^2} = \sqrt{4} = 2$

$x_3 = \sqrt{(1-1)^2+(1-1)^2+(0-3)^2} = \sqrt{9} = 3$

$x_4 = \sqrt{(1-1)^2+(1-2)^2+(0-2)^2} = \sqrt{1+1+4} = \sqrt{6} = 2.25$

$x_5 = \sqrt{(1-1)^2+(1-1)^2+(0-0)^2} = \sqrt{0+0+0} = 0.$

$x_6 = \sqrt{(1-6)^2+(1-6)^2+(0-10)^2} = \sqrt{25+25+100} = \sqrt{150} = 12.5$

Classification : Predicted value is 1. (outlier because it is too high).

Regression : Predicted value is $\frac{1+1+2}{3} = \frac{4}{3} = 1.$

8. consider the data in Q7 above. use this data to predict the missing value in the pattern vector (100, –, 100) with k=1. Is there a problem with (100, –, 100)? what can you say about this pattern?

Sol:- (100, –, 100) is extremely far. KNN is sensitive to scale and outliers.

* Increasing 'k' makes the model more robust but may include noise.

9. Consider the following 4 two dimensional vectors. Let
X1 = (1, 100000) X2 = (2, 100000) X3 = (1, 200000) X4 = (2, 200000)

a) How do we use range scaling scheme in this example.

b) Normalize the data using standard scale.

Sol:- Range scaling means minmax scaling.

$$\boxed{\text{minmax } x' = \frac{x - \min}{\max - \min}}$$

a) we have to choose the columnwise first.

x_1	1	100000
x_2	2	100000
x_3	1	200000
x_4	2	200000

min = 1
max = 2.

$x_1' = \frac{1-1}{2-1} = 0$, $x_2 = \frac{2-1}{2-1} = 1$, $x_3 = \frac{1-1}{2-1} = 0$,

(Column 1)

$x_4 = \frac{2-1}{2-1} = 1$.

Column 2:- min = 100000, max = 200000.

$x' = \frac{100000-100000}{200000-100000} = 0$, $x_3' = \frac{200000-100000}{200000-100000} = 1$

$\boxed{\text{The range scaling is 0 to 1.}}$

b) Standard scalar nothing but z-score normalization.

$$\boxed{x' = \frac{x-\mu}{\sigma}}$$

$x_1 = 1$
$x_2 = 2$
$x_3 = 1$
$x_4 = 2$.

for column 1:-

$\mu = \frac{1+2+1+2}{4} = \frac{6}{4} = \frac{3}{2} = 1.5$

$\sigma = \sqrt{\frac{(1-1.5)^2+(2-1.5)^2+(1-1.5)^2+(2-1.5)^2}{4}}$

$= \sqrt{\frac{0.25+0.25+0.25+0.25}{4}} = 0.5$.

$x' = \frac{1-1.5}{0.5} = -1$, $x' = \frac{2-1.5}{0.5} = 1$, $x' = -1$, $x' = 1$

for column 2:-

$\mu = \frac{600000}{4} = 150000$

$\sigma = \sqrt{(100000-150000)^2+(100000-150000)^2+(200000-150000)^2 + (200000-150000)^2}$

$x' = -1 \quad x' = -1 \quad x' = 1 \quad x' = 1$

$\therefore$ $\boxed{\text{z-score normalization}}$ (standard scalar) range is $\boxed{-1 \text{ to } 1}$

normalization : reducing the data redundancy.

b) Show that feature Selection is a special case of feature extraction

Feature Selection :-

choose a subset of existing features without modifying them.

Feature Extraction :-

Create new features from existing ones.

Ex:- PCA (principle component analysis),
LDA (linear discriminate analysis).

Mean = 0 , S.D = 1 $\Rightarrow$ standardization.

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